

Hunt Feng

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Academic Work Employment

Research Assistant

University of Saskatchewan | 2021 - 2024

- Set up EDIPIC MHD (Magnetohydrodynamics) simulations on multi-node CPU clusters
- Optimized & benchmark WarpX PIC (Particle-In-Cell) simulations on GPU
- Reduced PIC simulation runtime by ~30% through CUDA kernel optimization
- Reduced 30 minutes per analysis by automating post-processing using Python

Research Assistant

Simon Fraser University | 2020 - 2020

- Investigated the well-posedness of the equations of motion of a disk in a gas which is governed by Linear Boltzmann Equation.

Research Assistant

SFU's Big Data Hub | 2019 - 2019

- (Project 1) Enabled auto-detection of handwritten surveys with 93% accuracy by developing an OpenCV algorithm to identify the selected option from multiple-choice questions.
- (Project 2) Increased mineral output by 21% by studying outbreaks pattern in the field using neural network with Tensorflow and optimizing the network parameters.

Non-academic Work Employment

Product Developer

Bitstrata Systems Inc. | 2024 - 2025

- Developed a micrometer level precision mm-wave distance measurement system using radar data, applying Kalman filtering for real-time signal smoothing and precision enhancement.
- Calibrated mm-wave radar measurements by correcting for waveguide-induced propagation delays, ensuring physical consistency through system-level modeling.
- Designed robust weighing system using load cells and accelerometers; applied Kalman and median filtering to mitigate sensor noise and reject outliers.

Full-stack Developer

Sophosia Software Inc. | 2021/01 - present

- Led a team of 3 through ideation to pre-launch using Agile
- Grew a community to 200 users, gained 150+ stars and 4.2k downloads
- Automated releases, saving 30 minutes per update via GitHub Actions CI/CD

Junior Computational Plasma Scientist

General Fusion | 2022 - 2022

- Enabled ~30% faster MHD (Magnetohydrodynamics) simulations by implementing a time-implicit scheme, supporting ~1.5 times longer time step compare to time-explicit schemes.
- Implemented adaptive time stepping for the time-implicit scheme to automatically maximize time step.

Computational Plasma Engineer Coop

General Fusion | 2021 - 2021

- Ensured positive density and pressure in MHD simulations by implementing a GPU-based Provably Positive Discontinuous Galerkin Method using Taichi Lang, a High Performance Computing Language with a Python interface.
- Implemented GPU-based multi-species Particle-In-Cell (PIC) code using Taichi Lang for studying the ionization process in plasma.

Projects

Semi-Lagrangian Vlasov Solver with Immersed Boundary (GitHub: HuntFeng/vlasolver)

- Enabled GPU-based Vlasov-Poisson solver with immersed boundary method
- Debugged using cuda-gdb (compiled with -g, -O0 flags)
- Profiled with Nsight System / Compute
- Improved performance 10% by resolving thread contention

Physics-Informed Neural Net for Landau Damping (GitHub: HuntFeng/landau_damping_PINN)

- Built a PINN in PyTorch to model Landau damping dynamics
- Trained on only 10% of PIC simulation data to reduce compute costs
- Achieved 99% energy prediction accuracy within first 2 unit times

Teaching Experience

Teaching Assistant

University of Saskatchewan | 2022 - 2024

- Taught tutorials for first year physics.
- Assisted professor in first year physics labs.

Volunteer Activities

Data Analyst

SFU Silicon Quantum Technology Lab | 2018 - 2018

- Analyze SQL data and optimized quantum chip component configuration by Python.
- Analyzed spectrum data of quantum chip components and extracted parameters.

Presentations

(Poster, APS DPP 2025) *Noise-Free Grid-based Vlasov Modeling of Plasma-Object Interactions via a Unified Ghost-Fluid Immersed-Boundary Method.*

(Poster, APS DPP 2023) *Stability of the Transonic Plasma Flow in the Magnetic Nozzle.*

(Poster, CUPC 2021). *GPU-based Provably Positive Discontinuous Galerkin method for Multidimensional Ideal Magnetohydrodynamics.*

Skills

Programming: Python (Scientific packages, PyTorch, ...), C/C++, CUDA, HIP (transferable from CUDA), Tachi Lang, Kokkos, Matlab, Lua, MPI/OpenMP, Typescript, Vue.js, (No)SQL

DevOps & Tools: Slurm, Docker, Apptainer, CMake, Nsight System/Compute, GDB, Git(Hub)

HPC/Cloud Platforms: Niagara, Narval, Mist, Plato, GCP, AWS EC2

Simulation Tools: WarpX, Bout++, EDIPIC, Fusion 360, Paraview

Knowledge: Numerical analysis, Sparse matrix, PIC, MHD, Machine Learning, PINN, Optimization
Language: English, Cantonese, Mandarin, French (beginner)

Education

Master of Science - Physics and Engineering Physics

University of Saskatchewan | 2022 - 2024

Thesis: Spectral Instability of Plasma Flow in Magnetic Nozzle

Bachelor's Honours - Mathematical Physics

Simon Fraser University | 2017 - 2022

Certificates

WHMIS 2015, Simon Fraser University, Burnaby, BC

Using Apptainer, Compute Ontario 2024 Summer School

GPU Programming, Compute Ontario 2024 Summer School